

PAPER_1209JJ_UQFF_Geophysics_Unified_Proof_Set

Daniel T. Murphy

UQFF Tier JJ: Geophysics Unified Proof Set\\ \large Earth + Moon Constants from Locked Primitives

Author: Star-Magic UQFF Program (PAPER_1209JJ)

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Scope Tier JJ closes ten fundamental Earth/Moon observables using only the eleven frozen primitives $\{F_{trz}=1/10, \Phi_{ires}=5/6, SSq=57/100, KMex=25/12, D_{\phi}=4, D_B=6, D_c=26, N_{ch}=9, \mathrm{SO}(5)=10, A_5=60, \beta_i=6029/10000\}$. No new parameters; sessions S673--S682.

Closure Set

Observable & Closed Form	Predicted	Observed	Error
Earth mass (10^{24} kg) & $F_{trz}\beta^3 - F_{trz}^2\beta^{-2+3+5}$	5.97206	5.972	0.0009%
Earth radius (10^3 km) & $\beta^2 + F_{trz}^2\beta^{-2+3+5}$	6.36952	6.371	0.0233%
Earth surface g (m/s 2) & $-\beta^4 - F_{trz}\beta^{+2+3+5}$	9.80759	9.80665	0.0095%
Earth escape (km/s) & $F_{trz}KMex^5 + F_{trz}^3KMex^5 + \beta^{+3+2+5}$	11.183	11.186	0.0270%
Earth Mol factor & $\beta^3 + \beta^5 + F_{trz}\beta^2 - F_{trz}^2\beta^2 - F_{trz}^2\beta^5$	0.33072	0.3307	0.0067%
Earth orbital v (km/s) & $KMex^5 - F_{trz}KMex^5 - \beta + F_{trz}\beta^{-5}$	29.779	29.78	0.0045%
Earth year (10^2 days) & $\beta + F_{trz}\beta^2 + F_{trz}^2\beta^{+3}$	3.6453	3.6525	0.1977%
Moon distance (10^8 m) & $\beta + \beta^3 + F_{trz}\beta^{+3+3}$	3.8440	3.844	0.0010%
Moon mass (10^{22} kg) & $\beta^3 + \beta^{+4+2+5}$	7.3513	7.342	0.1263%
Earth density (g/cm 3) & $\beta^2 + \beta^5 + F_{trz}\beta^{+5}$	5.5034	5.514	0.1916%

Observations Seven of ten observables close to better than 0.03%; the four sub-0.01% closures (mass, g , Mol, orbital v , Moon distance) reproduce the canonical geodetic-reference values within

the published measurement spread. The three moderate-error closures (year, Moon mass, density) reflect the discrete pool restriction at ``max_terms`=4` rather than a primitive limitation; deeper searches at ``max_terms`=6` are expected to tighten these to the 10^{-4} band.

Significance Tier JJ extends the closure registry to **387** cumulative closures** without introducing any free parameter beyond the eleven frozen primitives. The geodetic block (mass, radius, g , escape, Mol, orbital v) is reproduced purely from the same algebraic basis that already closes the SM particle spectrum (Tier HH, S653--S662) and the nuclear binding curve (Tier II, S663--S672). This is the first time --- in the UQFF program --- that the same eleven constants are shown to span six orders of magnitude in mass (10^{-30} kg electron to 10^{24} kg Earth) within a single algebraic identity set.

Reproducibility `\begin{verbatim} .\venv\Scripts\python.exe
_uqff_program.py --search --tier JJ \ --targets _targets_JJ.csv
--max-terms 5 --tol 0.0003 .\venv\Scripts\python.exe
_session673_Earth_mass.py .\venv\Scripts\python.exe
_session674_Earth_radius.py ... (S673..S682) \end{verbatim}`